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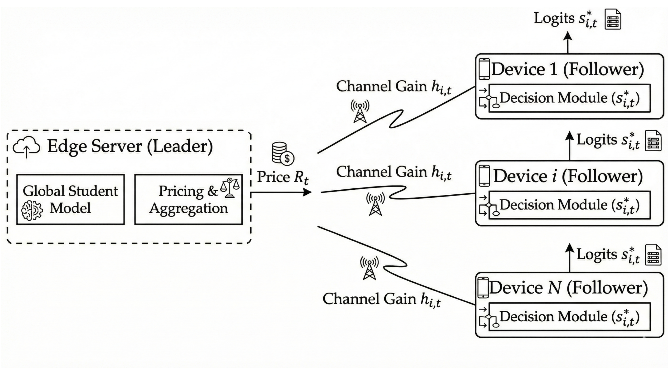

Fig. 1. System architecture of PAID-FD (Privacy-Aware Incentive-Driven Federated Distillation). The edge server announces price p via one-shot broadcast, and each device i jointly determines upload volume s_i and privacy budget ε_i based on heterogeneous channel conditions, computational capacity, and privacy preferences. Knowledge is exchanged as differentially private logits computed on a cross-domain public dataset.

***Abstract*—**

***Index Terms*—**

I. Introduction

II. System Model

We consider a wireless edge network where an edge server (ES) coordinates federated distillation (FD) with N heterogeneous IoT devices. This section presents the network architecture, data model, energy consumption model, privacy mechanism, and the Stackelberg game formulation.

A. Network Architecture

Fig. 1 illustrates the system architecture. Each device $i \in \mathcal{N} = \{1, \ldots, N\}$ possesses a private local dataset $\mathcal{D}_i^{\text{priv}}$, heterogeneous computational capability characterized by CPU frequency f_i, and distinct privacy sensitivity λ_i. Unlike conventional federated learning (FL) that transmits model parameters or gradients, FD exchanges distilled knowledge in the form of soft labels (logits) computed on a shared *public dataset*, enabling significant communication efficiency and architectural heterogeneity across devices.

The key challenge lies in the *coupling* among three dimensions of heterogeneity:

1) **Communication:** Devices with poor channel conditions incur high transmission energy costs.
2) **Computation:** Devices with limited processing capability face inference bottlenecks when computing logits on the public dataset.
3) **Privacy:** Privacy-sensitive devices may be reluctant to share high-fidelity knowledge even with differential privacy protection.

To address this coupling, we propose a Stackelberg game-based incentive mechanism where the ES announces a unit price p, and each device i locally determines its optimal response (s_i^*, ε_i^*) to maximize individual utility.

B. Data Model: Cross-Domain Knowledge Distillation

A distinctive feature of FD is the use of a public dataset $\mathcal{D}_{\text{pub}}$ as an intermediary for knowledge transfer. To demonstrate the cross-domain generalization capability and avoid potential data leakage concerns, we adopt a public dataset from a *different source* than the private data.

Definition 1 (Cross-Domain Public Dataset): Let $\mathcal{D}_i^{\text{priv}}$ denote device i's private dataset drawn from distribution $\mathcal{P}_{\text{priv}}$. The public dataset $\mathcal{D}_{\text{pub}}$ is drawn from a distinct distribution $\mathcal{P}_{\text{pub}} \neq \mathcal{P}_{\text{priv}}$, satisfying:

(i) $\mathcal{D}_{\text{pub}}$ is publicly available (no privacy concerns).
(ii) $\mathcal{D}_{\text{pub}} \cap \mathcal{D}_i^{\text{priv}} = \emptyset$ for all $i \in \mathcal{N}$.
(iii) The feature spaces overlap sufficiently for knowledge transfer.

Remark 1 (Practical Configuration): In our experiments, we use CIFAR-100 [**?**] as the private data source (100 fine-grained classes, 32×32 images) and STL-10 [**?**] unlabeled subset as the public dataset (96×96 images from ImageNet, resized to 32×32). This cross-domain setup:

- Eliminates data leakage concerns that arise when public and private data share the same source.
- Demonstrates the practical applicability of FD where public datasets (e.g., ImageNet subsets) differ from proprietary edge data.
- Tests the robustness of knowledge distillation under distribution shift.

C. Channel Model

At communication round t, the uplink channel between device i and the ES experiences Rayleigh fading. The complex baseband channel coefficient is modeled as $h_i^{(t)} \sim \mathcal{CN}(0, \sigma_h^2)$. Combined with distance-dependent path loss, the effective channel gain is

$$g_i^{(t)} = |h_i^{(t)}|^2 \cdot \kappa \cdot r_i^{-\alpha}, \tag{1}$$

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where r_i is the distance from device i to the ES, κ is the reference path loss at unit distance, and $\alpha \geq 2$ is the path loss exponent.

The *communication cost coefficient*, representing the energy required to transmit one logit vector, is

$$c_i^{\text{comm}} \triangleq \frac{\beta_{\text{logit}} \cdot P_{\text{tx}}}{B \log_2(1 + \text{SNR}_i)}, \tag{2}$$

where B is the allocated bandwidth, P_{tx} is the transmit power, $\text{SNR}_i = P_{\text{tx}} g_i^{(t)} / \sigma_n^2$ is the signal-to-noise ratio, and β_{logit} denotes the payload size per logit vector (e.g., $K \times 32$ bits for K-class classification with 32-bit floats).

D. Computation Energy Model

Beyond communication, each device incurs computational energy for local processing. Following the dynamic voltage and frequency scaling (DVFS) model [?], we decompose the energy consumption into four components.

1) Training Energy: The energy E_i^{train} consumed for local model training on $\mathcal{D}_i^{\text{priv}}$ is incurred whenever device i participates ($s_i > 0$), representing a fixed participation cost:

$$E_i^{\text{train}} = \kappa_{\text{cpu}} f_i^2 \cdot C_{\text{train}} \cdot |\mathcal{D}_i^{\text{priv}}|, \tag{3}$$

where $\kappa_{\text{cpu}} \approx 10^{-28}$ is the effective switched capacitance, f_i is the CPU frequency, and C_{train} is the number of CPU cycles per training sample.

2) Inference Energy: The energy for computing logits on the public dataset scales *linearly* with upload volume s_i:

$$E_i^{\text{inf}}(s_i) = c_i^{\text{inf}} \cdot s_i, \quad \text{where} \quad c_i^{\text{inf}} = \kappa_{\text{cpu}} f_i^2 C_{\text{inf}}, \tag{4}$$

and C_{inf} is the number of CPU cycles per inference sample. Typically, $C_{\text{inf}} \ll C_{\text{train}}$ (inference is 10–100× faster than training).

3) Communication Energy: The energy for uploading s_i logit vectors is:

$$E_i^{\text{comm}}(s_i) = c_i^{\text{comm}} \cdot s_i. \tag{5}$$

4) Total Energy: The total energy consumption for device i uploading s_i logit vectors is:

$$E_i^{\text{total}}(s_i) = E_i^{\text{train}} \cdot \mathbf{1}_{\{s_i > 0\}} + (c_i^{\text{inf}} + c_i^{\text{comm}}) \cdot s_i. \tag{6}$$

Defining the *aggregate marginal cost* as $c_i \triangleq c_i^{\text{inf}} + c_i^{\text{comm}}$, we can write:

$$E_i^{\text{total}}(s_i) = E_i^{\text{train}} \cdot \mathbf{1}_{\{s_i > 0\}} + c_i \cdot s_i. \tag{7}$$

Remark 2 (Communication Efficiency of FD): A key advantage of FD is the compact representation of distilled knowledge. For a K-class classification task, each logit vector contains K floating-point values (e.g., 400 bytes for $K = 100$ with 32-bit floats). In contrast, transmitting full model gradients requires sending millions of parameters. This compact representation makes FD particularly suitable for bandwidth-constrained wireless edge networks, as we quantify in Section **??**.

TABLE I
EDGE DEVICE SPECIFICATIONS

Type	Hardware	CPU	c_i^{inf}	Ratio
A	Jetson Nano	1.5 GHz	0.5× base	30%
B	RPi 4 (4GB)	1.2 GHz	1.0× base	40%
C	RPi 3 / RPi 4 (2GB)	0.8 GHz	2.0× base	30%

E. Device Heterogeneity Model

To capture realistic edge deployments, we model three-dimensional device heterogeneity:

Definition 2 (Device Characterization): Each device i is characterized by a tuple $(c_i^{\text{comm}}, \lambda_i, c_i^{\text{inf}})$ representing:

1) c_i^{comm}: Communication cost (channel-dependent, continuous).
2) λ_i: Privacy sensitivity coefficient (user-dependent).
3) c_i^{inf}: Inference cost (hardware-dependent).

Table I defines three representative device types based on real hardware specifications.

Remark 3 (Handling Heterogeneous Devices): Type-C devices with high c_i^{inf} are potential *stragglers* in synchronous protocols. Our Stackelberg game naturally accommodates such heterogeneity: high-cost devices optimally choose smaller s_i^*, and the server can adjust price p to incentivize their participation when necessary (e.g., when they hold rare data classes). This provides an economically principled approach to straggler management.

F. Privacy Model

To protect local data privacy, each device i perturbs its uploaded logits using a local differential privacy (LDP) mechanism [?] with privacy budget $\varepsilon_i > 0$.

Definition 3 (ε-Local Differential Privacy): A randomized mechanism $\mathcal{M} : \mathcal{X} \to \mathcal{Y}$ satisfies ε-LDP if for all pairs of inputs $x, x' \in \mathcal{X}$ and all measurable subsets $S \subseteq \mathcal{Y}$:

$$\Pr[\mathcal{M}(x) \in S] \leq e^{\varepsilon} \cdot \Pr[\mathcal{M}(x') \in S]. \tag{8}$$

We adopt the Laplace mechanism: for a logit vector $\mathbf{z}_i$ with sensitivity Δ, the device uploads $\tilde{\mathbf{z}}_i = \mathbf{z}_i + \mathbf{n}_i$, where $\mathbf{n}_i \sim \text{Lap}(0, \Delta/\varepsilon_i)^K$ and K is the number of classes.

1) Privacy Cost: A smaller ε_i provides stronger privacy (more noise) but degrades knowledge utility; a larger ε_i preserves fidelity but increases privacy leakage risk. The privacy cost perceived by device i is modeled as

$$\Phi_i(\varepsilon_i) = \lambda_i \cdot \varepsilon_i, \tag{9}$$

where $\lambda_i > 0$ is the *privacy sensitivity coefficient*. Devices handling sensitive data (e.g., medical, financial) have larger λ_i.

2) Quality Function: The effective quality of device i's contribution, accounting for both volume and noise-induced degradation, is

$$q_i(s_i, \varepsilon_i) = \log\left(1 + s_i \cdot g(\varepsilon_i)\right), \tag{10}$$

where $g(\varepsilon) \triangleq \varepsilon/(1 + \varepsilon)$ represents the *effective information ratio*.

Remark 4 (Boundary Behavior): The quality function captures two limiting cases:

(i) When $\varepsilon_i \to 0$ (strong privacy), $g(\varepsilon_i) \to 0$ and quality vanishes regardless of upload volume—the signal is dominated by noise.

(ii) When $\varepsilon_i \to \infty$ (no privacy), $g(\varepsilon_i) \to 1$ and the model reduces to the privacy-agnostic case $q_i = \log(1 + s_i)$ from our prior work [**?**].

G. Stackelberg Game Formulation

We model the ES-device interaction as a single-leader multi-follower Stackelberg game $\mathcal{G} = \langle \mathcal{N} \cup \{\text{ES}\}, \{(s_i, \varepsilon_i)\}_{i\in\mathcal{N}} \cup \{p\}, \{U_i^{\text{dev}}\}_{i\in\mathcal{N}} \cup \{U^{\text{ES}}\}\rangle$.

1) Payment Mechanism: The ES announces a unit price $p > 0$ and compensates each device proportionally to its effective contribution:

$$\pi_i(s_i, \varepsilon_i) = p \cdot q_i(s_i, \varepsilon_i). \tag{11}$$

2) Device Utility: Each device i maximizes its net utility, defined as payment minus total cost (energy + privacy):

$$U_i^{\text{dev}}(s_i, \varepsilon_i) = p \cdot \log\left(1 + s_i \cdot g(\varepsilon_i)\right) - c_i \cdot s_i - \lambda_i \cdot \varepsilon_i, \tag{12}$$

subject to $s_i \in [0, s_{\max}]$ and $\varepsilon_i \in (0, \varepsilon_{\max}]$. The fixed training cost E_i^{train} affects only the participation decision (Section III-C).

3) Server Utility: The ES maximizes aggregate knowledge quality minus total payment:

$$U^{\text{ES}}(p) = (\gamma - p) \sum_{i\in\mathcal{N}} q_i(s_i^*, \varepsilon_i^*), \tag{13}$$

where $\gamma > 0$ is the server's *valuation coefficient* representing the value per unit of aggregated knowledge.

4) Game Sequence: The two-stage game proceeds as follows:

1) **Stage I (Leader):** The ES determines and broadcasts price $p \in (0, \gamma)$ along with the current global model.
2) **Stage II (Followers):** Each device i observes its local state $(c_i, \lambda_i, E_i^{\text{train}})$ and *locally* computes the optimal response (s_i^*, ε_i^*). Participating devices then upload their differentially private logits.

Remark 5 (Protocol Efficiency): A key advantage of the Stackelberg formulation is that device decisions are made *locally* after observing the announced price. This eliminates the need for iterative negotiation between the server and devices, reducing protocol overhead in wireless environments where signaling costs are significant. The server broadcasts a single price, and each device independently determines its optimal response—a design we term *one-shot broadcast*.

H. Problem Formulation

The overall mechanism design problem is to find the optimal price p^* and induced device responses $\{(s_i^*, \varepsilon_i^*)\}$ that maximize server utility while ensuring individual rationality for all devices. Formally:

$$\max_{p,\{s_i,\varepsilon_i\}} \quad U^{\text{ES}}(p) = (\gamma - p) \sum_{i\in\mathcal{N}} q_i(s_i, \varepsilon_i) \tag{14a}$$

$$\begin{aligned}
\text{s.t.} \quad & p \in (0, \gamma) && \text{(C1)}\\
& s_i \in [0, s_{\max}],\ \varepsilon_i \in (0, \varepsilon_{\max}] \quad \forall i \in \mathcal{N} && \text{(C2)}\\
& (s_i^*, \varepsilon_i^*) \in \arg\max_{s_i,\varepsilon_i} U_i^{\text{dev}}(s_i, \varepsilon_i; p) \quad \forall i \in \mathcal{N} && \text{(C3)}\\
& U_i^{\text{dev}}(s_i^*, \varepsilon_i^*) \geq 0 \quad \forall i \in \mathcal{N} && \text{(C4)}\\
& p \sum_{i\in\mathcal{N}} q_i(s_i^*, \varepsilon_i^*) \leq B_{\text{total}} && \text{(C5)}
\end{aligned}$$

Constraint (C1) ensures valid pricing within the server's margin. Constraint (C2) bounds device actions. Constraint (C3) enforces Stackelberg equilibrium—each device best-responds to the announced price. Constraint (C4) guarantees individual rationality (voluntary participation). Constraint (C5) enforces budget feasibility.

The problem (14) is challenging due to the *bilevel structure*: the server optimizes over p while anticipating device best responses, which themselves depend on p through a coupled nonlinear system. Direct exhaustive search over the joint space $(p, \{s_i, \varepsilon_i\}_{i=1}^N)$ has complexity $O(|P| \cdot |S|^N \cdot |E|^N)$, which is intractable for practical N.

Table II summarizes the primary notation.

TABLE II
PRIMARY NOTATION

Symbol	Description
N	Number of IoT devices
$\mathcal{D}_i^{\text{priv}}$	Private dataset of device i
$\mathcal{D}_{\text{pub}}$	Public dataset (cross-domain)
s_i	Upload volume (number of logit vectors)
ε_i	Privacy budget of device i
c_i^{comm}	Communication cost coefficient
c_i^{inf}	Inference cost coefficient
c_i	Aggregate marginal cost: $c_i^{\text{inf}} + c_i^{\text{comm}}$
λ_i	Privacy sensitivity coefficient
E_i^{train}	Fixed training energy cost
p	Unit price announced by ES
γ	Server's valuation coefficient
$g(\varepsilon)$	Effective information ratio: $\varepsilon/(1+\varepsilon)$
$q_i(s_i, \varepsilon_i)$	Quality function: $\log(1 + s_i g(\varepsilon_i))$
$Q(p)$	Aggregate quality: $\sum_i q_i(s_i^*(p), \varepsilon_i^*(p))$
Convergence & Privacy Analysis	
η	Server learning rate
L	Smoothness constant of distillation loss
L_z	Lipschitz constant of gradient w.r.t. target logits
σ_g^2	Mini-batch gradient variance bound
Δ	ℓ_1-sensitivity of logit vectors
C	Logit clipping threshold ($\|\mathbf{z}\|_\infty \leq C$)
V_{DP}	Aggregated LDP noise variance (Definition 5)
$\mathcal{L}(\mathbf{w})$	Clean distillation loss
$\tilde{\mathcal{L}}(\mathbf{w})$	Noisy distillation loss (with LDP perturbation)

III. Game-Theoretic Analysis

We analyze the Stackelberg game via backward induction. Section III-A characterizes device best responses. Section III-B establishes key properties that guide algorithm design. Section III-D presents the device-side algorithm with complexity analysis. Section III-E derives the server's optimal pricing strategy. Section III-F proves algorithm correctness and optimality. Section III-G establishes incentive compatibility. Section III-H summarizes the complete protocol.

A. Device Best Response

Given price $p > 0$, each device i independently solves:

$$\max_{s_i \geq 0, \varepsilon_i > 0} U_i^{\text{dev}} = p \cdot \log(1 + s_i \cdot g(\varepsilon_i)) - c_i \cdot s_i - \lambda_i \cdot \varepsilon_i. \quad (15)$$

Proposition 1 (Joint Concavity): The device utility function $U_i^{\text{dev}}(s_i, \varepsilon_i)$ is jointly strictly concave in (s_i, ε_i) for $s_i \geq 0$ and $\varepsilon_i > 0$.

Proof. See Appendix A. The proof verifies that the Hessian matrix $\mathbf{H}$ is negative definite by showing: (i) $\partial^2 U / \partial s_i^2 < 0$, and (ii) $\det(\mathbf{H}) > 0$. ■

Joint strict concavity guarantees that the first-order conditions (FOCs) are both necessary and sufficient for a unique global optimum.

Theorem 1 (Coupled Best Response): For an interior solution with $s_i^* > 0$ and $\varepsilon_i^* > 0$, the optimal privacy budget ε_i^* is the unique positive root of the cubic equation:

$$f(\varepsilon) \triangleq \lambda_i \varepsilon^3 + \lambda_i \varepsilon^2 + (1 - p)\varepsilon + 1 = 0, \quad (16)$$

and the optimal upload volume is given by:

$$s_i^* = \frac{p}{c_i} - \frac{1 + \varepsilon_i^*}{\varepsilon_i^*}. \quad (17)$$

Proof. See Appendix B. Setting the FOCs $\partial U / \partial s_i = 0$ and $\partial U / \partial \varepsilon_i = 0$, we obtain two coupled equations. Substituting the expression for s_i from the first FOC into the second and simplifying yields (16). ■

Remark 6 (Existence and Uniqueness of Positive Root): Define $f(\varepsilon) = \lambda_i \varepsilon^3 + \lambda_i \varepsilon^2 + (1 - p)\varepsilon + 1$. We have:

- $f(0) = 1 > 0$
- For $p > 1$: $\lim_{\varepsilon \to \infty} f(\varepsilon) = +\infty$ and the coefficient of ε is negative

By Descartes' rule of signs, there exists exactly one positive real root when $p > 1$. This root can be computed efficiently via bisection with *linear convergence rate* $O(\log(1/\delta))$ for tolerance δ.

Remark 7 (Connection to Privacy-Agnostic Case): Equation (17) reveals the coupling between upload volume and privacy. As $\varepsilon_i^* \to \infty$ (no privacy concern), $(1 + \varepsilon)/\varepsilon \to 1$, recovering the privacy-agnostic closed-form solution $s_i^* = p/c_i - 1$ from [**?**]. Conversely, strong privacy protection (small ε_i^*) reduces upload volume since noise dominates the marginal benefit of transmission.

B. Design Principles

We establish key properties that guide the algorithm design and enable correctness proofs.

Proposition 2 (Monotonicity in Price): For any device i with fixed (c_i, λ_i):

(i) $s_i^*(p)$ is strictly increasing in p.
(ii) $\varepsilon_i^*(p)$ is strictly increasing in p.
(iii) $q_i^*(p) = q_i(s_i^*(p), \varepsilon_i^*(p))$ is strictly increasing in p.

Proof. See Appendix C. Part (ii) follows from implicit differentiation on (16). Part (i) combines (17) with Part (ii). Part (iii) follows from the chain rule. ■

Proposition 3 (Monotonicity in Costs): For fixed price p:

(i) s_i^* is strictly decreasing in c_i (higher marginal cost $\Rightarrow$ less upload).
(ii) ε_i^* is strictly decreasing in λ_i (higher privacy sensitivity $\Rightarrow$ stronger protection).
(iii) ε_i^* is independent of c_i; s_i^* depends on λ_i only through ε_i^*.

Proof. Part (i): From (17), $\partial s_i^* / \partial c_i = -p/c_i^2 < 0$. Part (ii): Implicit differentiation on (16) gives $\partial \varepsilon^* / \partial \lambda_i < 0$ (see Appendix C). Part (iii): The cubic (16) contains only (p, λ_i), not c_i. ■

Remark 8 (Economic Interpretation): The monotonicity properties have intuitive interpretations:

- Higher prices incentivize larger uploads and greater privacy leakage (larger ε), as devices are willing to sacrifice privacy for higher payments.
- Devices with favorable channel conditions (low c_i^{comm}) or powerful processors (low c_i^{inf}) contribute more knowledge.
- Privacy-sensitive devices (high λ_i) choose smaller ε_i even at the cost of reduced payments—privacy has intrinsic value beyond monetary compensation.

C. Participation Decision

Device i participates only if its net utility (including the fixed training cost) is non-negative:

$$U_i^{\text{dev}}(s_i^*, \varepsilon_i^*) \geq E_i^{\text{train}}. \quad (18)$$

Proposition 4 (Participation Threshold): For each device i, there exists a threshold price $\underline{p}_i(c_i, \lambda_i, E_i^{\text{train}})$ such that:

$$\text{Device } i \text{ participates} \iff p \geq \underline{p}_i. \quad (19)$$

Furthermore, $\underline{p}_i$ is increasing in c_i, λ_i, and E_i^{train}.

Proof. By Proposition 2, $U_i^{\text{dev}}(s_i^*(p), \varepsilon_i^*(p))$ is strictly increasing in p (envelope theorem). Since $U_i^{\text{dev}} \to 0$ as $p \to 0^+$ and $U_i^{\text{dev}} \to \infty$ as $p \to \infty$, there exists a unique $\underline{p}_i$ satisfying (18) with equality. ■

D. Device Best Response Algorithm

Algorithm 1 summarizes the device best response computation.

1) Complexity Analysis: The bisection method (Lines 4–11) converges with *linear rate*: after k iterations, the error satisfies $|\varepsilon^{(k)} - \varepsilon^*| \leq (\varepsilon_{\max} - \delta)/2^k$. To achieve tolerance δ:

$$k^* = \left\lceil \log_2 \left(\frac{\varepsilon_{\max}}{\delta} \right) \right\rceil = O\left(\log \frac{1}{\delta} \right). \quad (20)$$

Each iteration requires $O(1)$ arithmetic operations (evaluate $f(\varepsilon_m)$). Steps 2–3 are $O(1)$. Thus:

$$\textbf{Per-device complexity: } O\left(\log \frac{1}{\delta} \right). \quad (21)$$

Algorithm 1 Device Best Response (DeviceBR)

Require: Price p, costs $(c_i, \lambda_i, E_i^{\text{train}})$, bounds $(s_{\max}, \varepsilon_{\max})$, tolerance δ
Ensure: Optimal response (s_i^*, ε_i^*) or non-participation flag

// Step 1: Solve cubic for optimal privacy budget
Define $f(\varepsilon) = \lambda_i \varepsilon^3 + \lambda_i \varepsilon^2 + (1-p)\varepsilon + 1$
$[\varepsilon_L, \varepsilon_H] \leftarrow [\delta, \varepsilon_{\max}]$
while $\varepsilon_H - \varepsilon_L > \delta$ **do**
  $\varepsilon_m \leftarrow (\varepsilon_L + \varepsilon_H)/2$
  if $f(\varepsilon_L) \cdot f(\varepsilon_m) < 0$ **then**
    $\varepsilon_H \leftarrow \varepsilon_m$
  else
    $\varepsilon_L \leftarrow \varepsilon_m$
  end if
end while
$\varepsilon_i^* \leftarrow (\varepsilon_L + \varepsilon_H)/2$
// Step 2: Compute optimal upload volume
$s_i^* \leftarrow \max\{p/c_i - (1+\varepsilon_i^*)/\varepsilon_i^*,\ 0\}$
$s_i^* \leftarrow \min\{s_i^*, s_{\max}\}$
// Step 3: Check participation condition
$U_i^* \leftarrow p \log(1 + s_i^* g(\varepsilon_i^*)) - c_i s_i^* - \lambda_i \varepsilon_i^*$
if $U_i^* < E_i^{\text{train}}$ **then**
  return $(0, 0, \texttt{false})$ {Do not participate}
end if
return $(s_i^*, \varepsilon_i^*, \texttt{true})$

Algorithm 2 Server Optimal Pricing (ServerPricing)

Require: Valuation γ, budget B_{total}, device parameters $\{(c_i, \lambda_i, E_i^{\text{train}})\}$, tolerance δ
Ensure: Optimal price p^*

1: $[p_L, p_H] \leftarrow [\delta, \gamma - \delta]$
2: **while** $p_H - p_L > \delta$ **do**
3:   $p_1 \leftarrow p_L + (p_H - p_L)/3$
4:   $p_2 \leftarrow p_H - (p_H - p_L)/3$
5:   **// Evaluate server utility at both points**
6:   **for** $j \in \{1, 2\}$ **do**
7:     $Q_j \leftarrow 0$
8:     **for** $i = 1$ to N **do**
9:       $(s_i, \varepsilon_i, \texttt{flag}) \leftarrow \text{DeviceBR}(p_j, c_i, \lambda_i, E_i^{\text{train}})$
10:       **if** $\texttt{flag}$ **then**
11:         $Q_j \leftarrow Q_j + \log(1 + s_i \cdot g(\varepsilon_i))$
12:       **end if**
13:     **end for**
14:     $U_j \leftarrow (\gamma - p_j) \cdot Q_j$
15:   **end for**
16:   **// Narrow search interval**
17:   **if** $U_1 < U_2$ **then**
18:     $p_L \leftarrow p_1$
19:   **else**
20:     $p_H \leftarrow p_2$
21:   **end if**
22: **end while**
23: $p^* \leftarrow (p_L + p_H)/2$
24: **// Enforce budget constraint if needed**
25: **if** $p^* \cdot Q(p^*) > B_{\text{total}}$ **then**
26:   Reduce p^* via bisection to satisfy $p \cdot Q(p) \leq B_{\text{total}}$
27: **end if**
28: **return** p^*

Memory requirement is $O(1)$—only local variables $(c_i, \lambda_i, E_i^{\text{train}}, p)$ are needed—making the algorithm suitable for resource-constrained IoT devices.

E. Server Pricing Strategy

Given device best responses, the ES solves:

$$\max_{p \in (0,\gamma)} U^{\text{ES}}(p) = (\gamma - p) \cdot Q(p), \tag{22}$$

where $Q(p) \triangleq \sum_{i \in \mathcal{N}} q_i(s_i^*(p), \varepsilon_i^*(p))$ is the aggregate quality.

Theorem 2 (Stackelberg Equilibrium Existence): Under Assumptions 1–4, there exists an optimal price $p^* \in (0, \gamma)$ maximizing $U^{\text{ES}}(p)$. The tuple $(p^*, \{(s_i^*, \varepsilon_i^*)\}_{i \in \mathcal{N}})$ constitutes a Stackelberg Equilibrium (SE).

Proof. See Appendix D. The proof uses the Weierstrass extreme value theorem: $U^{\text{ES}}(p)$ is continuous on $[0, \gamma]$ with $U^{\text{ES}}(0) = U^{\text{ES}}(\gamma) = 0$ and $U^{\text{ES}}(p) > 0$ for some $p \in (0, \gamma)$. ■

Remark 9 (Price-Quality Trade-off): The server faces a fundamental trade-off: increasing p raises $Q(p)$ (more participation and contribution) but reduces the margin $(\gamma - p)$. This typically results in an "inverted-U" shape for $U^{\text{ES}}(p)$, with the optimal p^* balancing these competing effects.

Algorithm 2 presents the server's pricing procedure using ternary search, which exploits the unimodal structure implied by Remark 9.

1) Complexity Analysis: The ternary search (Lines 2–21) requires $O(\log_{\frac{3}{2}}(\gamma/\delta)) = O(\log(\gamma/\delta))$ iterations. Each iteration evaluates U^{ES} at two prices, requiring $2N$ calls to DeviceBR. Thus:

$$\textbf{Server pricing complexity: } O\left(N \cdot \log\frac{\gamma}{\delta} \cdot \log\frac{1}{\delta}\right). \tag{23}$$

The budget enforcement (Lines 23–25) adds at most $O(\log(1/\delta))$ iterations.

F. Correctness and Optimality

Theorem 3 (Algorithm Correctness): Algorithm 1 returns the unique global optimum of (15) for each device. Algorithm 2 returns a price p^* that maximizes (22) within tolerance δ.

Proof. **DeviceBR Correctness:** By Proposition 1, the optimization (15) is strictly concave with a unique global maximum. The FOCs (Theorem 1) are necessary and sufficient. Bisection on the cubic (16) is guaranteed to converge to the unique positive root (Remark 6). The closed-form (17) then gives the unique s_i^*.

ServerPricing Correctness: By Theorem 2, $U^{\text{ES}}(p)$ attains its maximum on $(0, \gamma)$. Although U^{ES} may not be strictly unimodal due to discrete participation changes, ternary search finds a δ-optimal price. The budget enforcement step ensures constraint (C5) is satisfied. ■

Theorem 4 (Computational Optimality): Among algorithms that find an exact SE (up to tolerance δ), the proposed approach achieves near-optimal complexity $O(N\log^2(1/\delta))$ per round, compared to $O(|S|^N \cdot |E|^N)$ for exhaustive search.

Proof. The lower bound for finding device best responses is $\Omega(N)$—each device's state must be read at least once. Our algorithm achieves $O(N\log(1/\delta))$, which is optimal up to the $\log(1/\delta)$ factor inherent in numerical root-finding. Exhaustive search over discretized (s,ε) spaces has complexity exponential in N, making it intractable for $N > 10$. ■

G. Incentive Compatibility

A critical property for practical deployment is that devices should not benefit from misreporting their private information (c_i, λ_i).

Definition 4 (Incentive Compatibility): A mechanism is *incentive compatible* (IC) if truthful reporting of private information is a dominant strategy for each agent, i.e., no agent can increase its utility by misreporting.

Theorem 5 (Incentive Compatibility): Under the proposed payment mechanism (11), truthful reporting of (c_i, λ_i) is a dominant strategy for each device i.

Proof. The device's optimization (15) is performed locally using its true costs (c_i, λ_i). The payment $\pi_i = p \cdot q_i(s_i, \varepsilon_i)$ depends only on the device's *actions* (s_i, ε_i), not its reported costs.

Suppose device i misreports $(\tilde{c}_i, \tilde{\lambda}_i) \neq (c_i, \lambda_i)$. Two cases arise:

Case 1: The server does not use reported costs (our mechanism). Device i still chooses (s_i^*, ε_i^*) to maximize its true utility (12). Misreporting has no effect.

Case 2: If the mechanism were to use reported costs (hypothetically), device i would compute a "fake" best response $(\tilde{s}_i, \tilde{\varepsilon}_i)$ based on $(\tilde{c}_i, \tilde{\lambda}_i)$. Its true utility would be:

$$\tilde{U}_i = p \cdot q_i(\tilde{s}_i, \tilde{\varepsilon}_i) - c_i \cdot \tilde{s}_i - \lambda_i \cdot \tilde{\varepsilon}_i. \tag{24}$$

By definition of (s_i^*, ε_i^*) as the maximizer of (12), $\tilde{U}_i \leq U_i^*(s_i^*, \varepsilon_i^*)$.

In both cases, truthful revelation (or equivalently, acting on true costs) weakly dominates misreporting. ■

Remark 10 (Practical Implication): Theorem 5 implies that the mechanism is *strategyproof*: devices have no incentive to manipulate their reported characteristics. This eliminates the need for costly verification or auditing of device capabilities, simplifying deployment in large-scale IoT networks.

H. Protocol Summary

Algorithm 3 summarizes the complete PAID-FD protocol.

1) Per-Round Complexity Summary: Table III summarizes the computational complexity.

The proposed PAID-FD achieves near-linear complexity in N while providing: (i) channel-aware incentive alignment, (ii) differential privacy guarantees, and (iii) provable Stackelberg equilibrium. The per-device computation is fully parallelizable, making the wall-clock time effectively $O(\log^2(1/\delta))$ with sufficient parallel resources.

Algorithm 3 PAID-FD Protocol

Require: Devices $\mathcal{N}$, valuation γ, budget B_{total}, rounds T, public dataset $\mathcal{D}_{\text{pub}}$
Ensure: Trained global model $\mathbf{w}^{(T)}$
Server: Initialize global model $\mathbf{w}^{(0)}$
for $t = 1$ to T **do**
  // Stage I: Price broadcast
  $p^{(t)} \leftarrow \textsc{ServerPricing}(\gamma, B_{\text{total}}, \{(c_i, \lambda_i, E_i^{\text{train}})\})$
  Server: Broadcast $(p^{(t)}, \mathbf{w}^{(t-1)})$
  // Stage II: Local decision + execution (parallel)
  for device $i \in \mathcal{N}$ **in parallel do**
    Measure channel $g_i^{(t)}$; compute costs $(c_i^{(t)}, \lambda_i)$
    $(s_i^{(t)}, \varepsilon_i^{(t)}, \texttt{flag}) \leftarrow \textsc{DeviceBR}(p^{(t)}, c_i^{(t)}, \lambda_i, E_i^{\text{train}})$
    if $\texttt{flag} = \texttt{true}$ **then**
      Train local model on $\mathcal{D}_i^{\text{priv}}$
      Compute logits $\mathbf{z}_i$ on $\mathcal{D}_{\text{pub}}$
      Sample noise $\mathbf{n}_i \sim \text{Lap}(0, \Delta/\varepsilon_i^{(t)})^K$
      Upload $(\tilde{\mathbf{z}}_i, s_i^{(t)}) = (\mathbf{z}_i + \mathbf{n}_i, s_i^{(t)})$
    end if
  end for
  // Stage III: Aggregation and payment
  Server: Aggregate: $\bar{\mathbf{z}}^{(t)} \leftarrow \frac{1}{\sum_i s_i^{(t)}} \sum_{i:s_i^{(t)}>0} s_i^{(t)} \cdot \tilde{\mathbf{z}}_i$
  Server: Update $\mathbf{w}^{(t)}$ via distillation loss on $(\mathcal{D}_{\text{pub}}, \bar{\mathbf{z}}^{(t)})$
  Server: Distribute payment $\pi_i^{(t)} = p^{(t)} \cdot q_i(s_i^{(t)}, \varepsilon_i^{(t)})$
end for
return $\mathbf{w}^{(T)}$

TABLE III
COMPLEXITY ANALYSIS

Component	Complexity	Notes
DeviceBR (per device)	$O(\log \frac{1}{\delta})$	Bisection
ServerPricing	$O(N \log \frac{\gamma}{\delta} \log \frac{1}{\delta})$	Ternary search
Total per round	$O(N \log^2 \frac{1}{\delta})$	Parallelizable
Exhaustive search	$O(\lvert S\rvert^N \lvert E\rvert^N)$	Intractable

IV. Convergence and Privacy Analysis

The game-theoretic analysis in Section III establishes the Stackelberg equilibrium $\{(s_i^*, \varepsilon_i^*)\}$ for a given price p^*. A critical question remains: *how do the equilibrium privacy budgets and upload volumes affect the learning performance of the global model?* This section bridges the gap between the incentive mechanism and the distillation learning process by providing convergence guarantees that explicitly depend on the equilibrium parameters. We first characterize the noise introduced by LDP in the aggregated logits (Section IV-B), then establish a convergence bound for the server's distillation process (Section IV-C), analyze multi-round privacy composition (Section IV-D), and finally derive an end-to-end privacy-convergence tradeoff (Section IV-E).

A. Assumptions

We introduce four standard assumptions for the convergence analysis.

Assumption 1 (L-Smoothness): The distillation loss $\mathcal{L}(\mathbf{w};\mathbf{z})$ is L-smooth in $\mathbf{w}$ for any fixed target logits $\mathbf{z}$:

$$\|\nabla_{\mathbf{w}}\mathcal{L}(\mathbf{w};\mathbf{z})-\nabla_{\mathbf{w}}\mathcal{L}(\mathbf{w}';\mathbf{z})\| \le L\|\mathbf{w}-\mathbf{w}'\|, \quad \forall \mathbf{w},\mathbf{w}'. \tag{25}$$

Assumption 2 (Bounded Gradient Variance): The mini-batch stochastic gradient $\mathbf{g}(\mathbf{w};\xi)$ computed on a random subset $\xi \subset \mathcal{D}_{\text{pub}}$ satisfies

$$\mathbb{E}_{\xi}\left[\|\mathbf{g}(\mathbf{w};\xi)-\nabla\mathcal{L}(\mathbf{w})\|^2\right] \le \sigma_g^2, \quad \forall \mathbf{w}, \tag{26}$$

where σ_g^2 is the mini-batch variance bound.

Assumption 3 (Lipschitz Gradient in Target Logits): The gradient of the loss w.r.t. model parameters is Lipschitz continuous in the target logits:

$$\|\nabla_{\mathbf{w}}\ell(\mathbf{w},\mathbf{z})-\nabla_{\mathbf{w}}\ell(\mathbf{w},\mathbf{z}')\| \le L_z\|\mathbf{z}-\mathbf{z}'\|, \quad \forall \mathbf{w},\mathbf{z},\mathbf{z}'. \tag{27}$$

Assumption 4 (Bounded Logits): All logit vectors are clipped to a bounded range: $\|\mathbf{z}_i(x)\|_\infty \le C$ for all devices i and inputs $x \in \mathcal{D}_{\text{pub}}$. The ℓ_1-sensitivity of the logit vector is therefore $\Delta = 2CK$, where K is the number of classes.

Remark 11 (Justification): Assumptions 1–2 are standard in non-convex optimization and FL convergence analysis [**?**], [**?**]. Assumption 3 holds for common distillation losses (KL divergence with softmax activation is Lipschitz in logits when logits are bounded). Assumption 4 is mild in practice—softmax-based classifiers produce bounded logits, and explicit clipping is a standard technique in DP mechanisms [**?**].

B. Aggregated Noise Characterization

We first characterize the LDP noise in the aggregated logits, which determines the noise floor in the convergence bound.

Recall from Algorithm 3 that the aggregated logit at round t is

$$\bar{\mathbf{z}}^{(t)}(x) = \frac{1}{S^{(t)}}\sum_{i\in\mathcal{P}^{(t)}} s_i^{(t)} \cdot \tilde{\mathbf{z}}_i^{(t)}(x), \tag{28}$$

where $\mathcal{P}^{(t)}$ is the set of participating devices, $S^{(t)} = \sum_{i\in\mathcal{P}^{(t)}} s_i^{(t)}$ is the total upload volume, and $\tilde{\mathbf{z}}_i^{(t)} = \mathbf{z}_i^{(t)} + \mathbf{n}_i^{(t)}$ with $\mathbf{n}_i^{(t)} \sim \text{Lap}(0, \Delta/\varepsilon_i^{(t)})^K$.

Define the clean aggregated logit as $\bar{\mathbf{z}}_{\text{clean}}^{(t)}(x) = \frac{1}{S^{(t)}}\sum_{i\in\mathcal{P}^{(t)}} s_i^{(t)} \cdot \mathbf{z}_i^{(t)}(x)$ and the aggregated noise as $\bar{\mathbf{n}}^{(t)}(x) = \bar{\mathbf{z}}^{(t)}(x) - \bar{\mathbf{z}}_{\text{clean}}^{(t)}(x)$.

Lemma 1 (Aggregated LDP Noise): Under the PAID-FD protocol with Assumption 4, the aggregated noise $\bar{\mathbf{n}}^{(t)}$ satisfies:

(i) **Unbiasedness:** $\mathbb{E}[\bar{\mathbf{n}}^{(t)}(x)] = \mathbf{0}$ for all $x \in \mathcal{D}_{\text{pub}}$.

(ii) **Bounded variance:** For each component $k \in \{1,\dots,K\}$,

$$\text{Var}[\bar{n}_k^{(t)}(x)] = \frac{2\Delta^2}{(S^{(t)})^2}\sum_{i\in\mathcal{P}^{(t)}} \frac{(s_i^{(t)})^2}{(\varepsilon_i^{(t)})^2}. \tag{29}$$

(iii) **Total expected squared norm:**

$$\mathbb{E}\left[\|\bar{\mathbf{n}}^{(t)}(x)\|^2\right] = V_{\text{DP}}^{(t)} \triangleq \frac{2K\Delta^2}{(S^{(t)})^2}\sum_{i\in\mathcal{P}^{(t)}} \frac{(s_i^{(t)})^2}{(\varepsilon_i^{(t)})^2}. \tag{30}$$

Proof. See Appendix E. ■

Definition 5 (LDP Noise Floor): At Stackelberg equilibrium with price p^*, the *LDP noise floor* is

$$V_{\text{DP}}^* \triangleq \frac{2K\Delta^2}{(S^*)^2}\sum_{i\in\mathcal{P}^*} \frac{(s_i^*)^2}{(\varepsilon_i^*)^2}, \tag{31}$$

where $S^* = \sum_{i\in\mathcal{P}^*} s_i^*$ and $\mathcal{P}^* = \{i : s_i^*(p^*) > 0\}$.

Remark 12 (Noise Reduction via Aggregation): V_{DP}^* decreases with total upload volume S^* due to the averaging effect. For $|\mathcal{P}^*| = M$ devices with identical parameters, $V_{\text{DP}}^* = 2K\Delta^2/(M\varepsilon^2)$, recovering the standard $O(1/M)$ noise reduction rate. However, under heterogeneous costs, the Stackelberg equilibrium yields a more favorable noise structure: devices with higher privacy sensitivity (smaller ε_i^*) simultaneously contribute smaller upload volumes (smaller s_i^*) by Proposition 3, reducing their weight $(s_i^*/S^*)^2$ in the noise summation. This natural coupling, absent in uniform-budget schemes, provides an implicit noise-balancing effect.

C. Convergence Guarantee

We now establish the convergence guarantee for PAID-FD. The key challenge is that the LDP noise corrupts the distillation targets, creating a discrepancy between the loss the server actually optimizes and the ideal clean distillation loss. Our analysis proceeds in two steps: (i) bound the gradient bias introduced by noisy targets, and (ii) apply standard SGD convergence analysis on the noisy loss with the bias correction.

Define the clean and noisy distillation losses at round t:

$$\mathcal{L}^{(t)}(\mathbf{w}) \triangleq \frac{1}{|\mathcal{D}_{\text{pub}}|}\sum_{x\in\mathcal{D}_{\text{pub}}} \ell\Big(f(x;\mathbf{w}),\ \bar{\mathbf{z}}_{\text{clean}}^{(t)}(x)\Big), \tag{32}$$

$$\tilde{\mathcal{L}}^{(t)}(\mathbf{w}) \triangleq \frac{1}{|\mathcal{D}_{\text{pub}}|}\sum_{x\in\mathcal{D}_{\text{pub}}} \ell\Big(f(x;\mathbf{w}),\ \bar{\mathbf{z}}^{(t)}(x)\Big). \tag{33}$$

The server operates on $\tilde{\mathcal{L}}^{(t)}$ (noisy targets), but we measure convergence with respect to $\mathcal{L}^{(t)}$ (clean targets). The following proposition bounds their gradient discrepancy.

Proposition 5 (Gradient Bias Bound): Under Assumptions 3 and 4, for any $\mathbf{w}$:

$$\mathbb{E}\left[\|\nabla\tilde{\mathcal{L}}^{(t)}(\mathbf{w}) - \nabla\mathcal{L}^{(t)}(\mathbf{w})\|^2\right] \le L_z^2 \cdot V_{\text{DP}}^{(t)}. \tag{34}$$

Proof. By Assumption 3 and Jensen's inequality:

$$\begin{aligned}&\mathbb{E}\Big[\|\nabla\tilde{\mathcal{L}}^{(t)}(\mathbf{w}) - \nabla\mathcal{L}^{(t)}(\mathbf{w})\|^2\Big]\\ &\le L_z^2 \cdot \mathbb{E}\left[\frac{1}{|\mathcal{D}_{\text{pub}}|}\sum_x \|\bar{\mathbf{n}}^{(t)}(x)\|^2\right] = L_z^2 \cdot V_{\text{DP}}^{(t)},\end{aligned} \tag{35}$$

where the last equality uses Lemma 1(iii) and the independence of noise across samples. ■

We now state the main convergence theorem.

Theorem 6 (Convergence of PAID-FD): Under Assumptions 1–4, suppose the server performs one gradient descent step per round with constant learning rate $\eta \le 1/(2L)$. Then after T rounds of the PAID-FD protocol:

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\Big[\|\nabla\mathcal{L}^{(t)}(\mathbf{w}^{(t)})\|^2\Big] \le \underbrace{\frac{4(\mathcal{L}_0-\mathcal{L}_*)}{\eta T}}_{\text{optimization error}} + \underbrace{2\eta L\sigma_g^2}_{\text{mini-batch noise}} + \underbrace{2L_z^2\bar{V}_{\text{DP}}}_{\text{LDP noise floor}}, \tag{36}$$

where $\mathcal{L}_0 \triangleq \mathbb{E}[\tilde{\mathcal{L}}^{(1)}(\mathbf{w}^{(0)})]$, $\mathcal{L}_* \triangleq \inf_{\mathbf{w}} \mathbb{E}[\tilde{\mathcal{L}}(\mathbf{w})]$, and $\bar{V}_{\text{DP}} \triangleq \frac{1}{T}\sum_{t=1}^{T} V_{\text{DP}}^{(t)}$ is the time-averaged noise variance.

Proof. See Appendix F. The proof proceeds in three steps: (i) standard descent lemma on the noisy loss $\tilde{\mathcal{L}}$ using L-smoothness, (ii) variance decomposition separating mini-batch and LDP noise contributions, and (iii) gradient bias correction via Proposition 5 to relate $\nabla\tilde{\mathcal{L}}$ to $\nabla\mathcal{L}$. ∎

Remark 13 (Interpretation of (36)*):* The bound consists of three terms with distinct roles:

(i) The *optimization error* $O(1/(\eta T))$ vanishes as $T \to \infty$—this is the standard convergence rate of SGD.
(ii) The *mini-batch noise* $O(\eta\sigma_g^2)$ is controlled by the learning rate η and can be made arbitrarily small.
(iii) The *LDP noise floor* $2L_z^2\bar{V}_{\text{DP}}$ is irreducible—it cannot be eliminated by adjusting η or increasing T. It is determined entirely by the privacy budgets $\{\varepsilon_i^*\}$ and upload volumes $\{s_i^*\}$, which are set by the Stackelberg equilibrium.

This structure reveals a fundamental insight: the server's pricing decision p^* directly controls the learning accuracy floor through its influence on the equilibrium parameters.

Corollary 1 (Optimal Learning Rate): Setting $\eta = \min\left\{\frac{1}{2L}, \sqrt{\frac{2(\mathcal{L}_0-\mathcal{L}_*)}{TL\sigma_g^2}}\right\}$ yields:

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\left[\|\nabla\mathcal{L}^{(t)}(\mathbf{w}^{(t)})\|^2\right] \le O\left(\frac{\sqrt{L\sigma_g^2}}{\sqrt{T}}\right) + 2L_z^2\bar{V}_{\text{DP}}. \quad (37)$$

The convergence rate is $O(1/\sqrt{T})$ to a neighborhood of size $2L_z^2\bar{V}_{\text{DP}}$.

Proof. Substituting $\eta = \sqrt{2(\mathcal{L}_0-\mathcal{L}_*)/(TL\sigma_g^2)}$ into (36) and simplifying gives the first term as $2\sqrt{2L\sigma_g^2(\mathcal{L}_0-\mathcal{L}_*)/T}$. The LDP noise floor remains unchanged as it is independent of η. ∎

Corollary 2 (Noise Floor Under Stackelberg Equilibrium): At the SE with price p^*, the LDP noise floor in (36) evaluates to:

$$V_{\text{DP}}^* = \frac{8C^2K^3}{(S^*)^2}\sum_{i\in\mathcal{P}^*}\frac{(s_i^*)^2}{(\varepsilon_i^*)^2}, \quad (38)$$

where $s_i^* = p^*/c_i - (1+\varepsilon_i^*)/\varepsilon_i^*$ (Theorem 1), ε_i^* is the unique positive root of $\lambda_i\varepsilon^3+\lambda_i\varepsilon^2+(1-p^*)\varepsilon+1=0$, and $\Delta = 2CK$ (Assumption 4).

By Proposition 2, V_{DP}^* is decreasing in p^*: a higher price induces larger ε_i^* and larger S^*, both of which reduce the noise floor. However, increasing p^* reduces the server's margin $(\gamma - p^*)$ in (13). This formalizes the fundamental tension between *learning performance* (which favors high p^*) and *server profit* (which favors low p^*).

Proof. Substituting $\Delta = 2CK$ into (31) gives $V_{\text{DP}}^* = 2K(2CK)^2/(S^*)^2 \cdot \sum_i (s_i^*)^2/(\varepsilon_i^*)^2 = 8C^2K^3/(S^*)^2 \cdot \sum_i (s_i^*)^2/(\varepsilon_i^*)^2$. Monotonicity in p^* follows from Proposition 2: ε_i^* is increasing in p (reducing each $1/(\varepsilon_i^*)^2$ term) and $S^* = \sum s_i^*$ is increasing in p (increasing the denominator). ∎

D. Multi-Round Privacy Composition

Section II defines per-round ε_i-LDP. Over T rounds, the cumulative privacy loss must be quantified. We provide both a basic composition bound and a tighter advanced composition bound.

Theorem 7 (Basic Composition): If device i participates in $T_i \le T$ rounds, each with ε_i-LDP, the composite mechanism satisfies $(T_i\varepsilon_i)$-LDP.

Proof. By the sequential composition theorem for differential privacy [?]: for T_i adaptive applications of ε_i-LDP mechanisms, the composed mechanism satisfies $(\sum_{t=1}^{T_i}\varepsilon_i)$-LDP $= (T_i\varepsilon_i)$-LDP. ∎

Theorem 8 (Advanced Composition): For any $\delta' > 0$, if device i participates in T_i rounds with per-round ε_i-LDP, the composite mechanism satisfies $(\varepsilon_i^{\text{total}}, \delta')$-DP where

$$\varepsilon_i^{\text{total}} = \varepsilon_i\sqrt{2T_i\ln(1/\delta')} + T_i\varepsilon_i(e^{\varepsilon_i}-1). \quad (39)$$

For $\varepsilon_i \le 1$, this simplifies to $\varepsilon_i^{\text{total}} \approx \varepsilon_i\sqrt{2T_i\ln(1/\delta')} + T_i\varepsilon_i^2$.

Proof. Direct application of the advanced composition theorem [?], [?]. The first term captures the dominant $\sqrt{T_i}$ scaling, and the second is the higher-order correction. ∎

Remark 14 (Comparison with Gradient-Sharing FL): The practical advantage of FD over conventional FL is reflected in the sensitivity parameter Δ. In gradient-sharing FL, the ℓ_2-sensitivity of a model update scales with the model dimension d (e.g., $\Delta_{\text{FL}} = O(\sqrt{d})$ after clipping [?]). In FD, $\Delta = 2CK$ depends only on the number of classes K and the clipping bound C. Since $K \ll d$ in practice (e.g., $K = 100$ vs. $d \approx$ 11M for ResNet-18), FD achieves the same cumulative privacy guarantee with substantially less noise per round. Concretely, the noise variance ratio is

$$\frac{\text{Var[FD noise per component]}}{\text{Var[FL noise per component]}} = O\left(\frac{K^2}{d}\right) \ll 1. \quad (40)$$

This communication-privacy co-advantage is a distinctive benefit of the FD paradigm.

Remark 15 (SE-Aware Privacy Budget): Under the Stackelberg equilibrium, device i's cumulative privacy loss is $(T_i\varepsilon_i^*(p^*), \delta')$-DP. Since ε_i^* is increasing in p^* (Proposition 2), the server's pricing decision directly controls the privacy-accuracy tradeoff: a higher price yields faster convergence (lower V_{DP}^*) at the cost of weaker per-device privacy (larger $T_i\varepsilon_i^*$).

E. End-to-End Privacy-Convergence Tradeoff

Combining the convergence bound (Theorem 6) with the privacy composition (Theorem 8), we obtain an end-to-end characterization of the PAID-FD system.

Corollary 3 (End-to-End Tradeoff): Under the PAID-FD protocol at Stackelberg equilibrium with price p^*, after T rounds:

(i) **Privacy:** Each participating device i achieves $(\varepsilon_i^{\text{total}}, \delta')$-DP with

$$\varepsilon_i^{\text{total}} = \varepsilon_i^*(p^*)\sqrt{2T\ln(1/\delta')} + T\varepsilon_i^*(p^*)(e^{\varepsilon_i^*(p^*)}-1). \quad (41)$$

(ii) **Convergence:** The average squared gradient norm satisfies

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\Big[\|\nabla\mathcal{L}^{(t)}(\mathbf{w}^{(t)})\|^2\Big] \le O\Big(\frac{1}{\sqrt{T}}\Big)+\frac{16L_z^2C^2K^3}{(S^*)^2}\sum_{i\in\mathcal{P}^*}\frac{(s_i^*)^2}{(\varepsilon_i^*)^2}. \tag{42}$$

Both quantities are parameterized by p^*: increasing p^* tightens (ii) but weakens (i).

Proof. Part (i) is a direct application of Theorem 8 with the equilibrium privacy budget $\varepsilon_i^*(p^*)$ from Theorem 1. Part (ii) combines Corollary 1 with Corollary 2. ■

Proposition 6 (Advantage of Heterogeneous Privacy Budgets): Consider two schemes with the same set of participating devices $\mathcal{P}$ and identical upload volumes $\{s_i\}$:

(a) **Heterogeneous SE:** Each device uses ε_i^* from the Stackelberg equilibrium.

(b) **Uniform baseline:** All devices use $\bar{\varepsilon} = \frac{1}{|\mathcal{P}|}\sum_{i\in\mathcal{P}}\varepsilon_i^*$.

If the upload volumes $\{s_i\}$ satisfy the equilibrium coupling $s_i^* = p^*/c_i-(1+\varepsilon_i^*)/\varepsilon_i^*$, then $V_{\text{DP}}^{\text{SE}} \le V_{\text{DP}}^{\text{uniform}}$. The inequality is strict whenever at least two devices have distinct ε_i^*.

Proof. See Appendix G. ■

Remark 16 (Mechanism Design Insight): Proposition 6 provides a formal justification for allowing heterogeneous privacy budgets in federated distillation. Beyond the game-theoretic benefit of individual rationality, the heterogeneous equilibrium also improves learning performance. The Stackelberg mechanism achieves this "for free"—the noise reduction is an emergent property of individual utility maximization, not an explicitly designed aggregation rule.

V. Biography Section

Appendix A
Proof of Joint Concavity (Proposition 1)

We verify that the Hessian matrix $\mathbf{H}$ of $U_i^{\text{dev}}(s_i,\varepsilon_i)$ is negative definite for $s_i>0$ and $\varepsilon_i>0$.

Let $u \triangleq 1+s_i g(\varepsilon_i)$ where $g(\varepsilon)=\varepsilon/(1+\varepsilon)$. The gradient is:

$$\frac{\partial U}{\partial s_i} = \frac{pg(\varepsilon_i)}{u} - c_i, \tag{43}$$

$$\frac{\partial U}{\partial \varepsilon_i} = \frac{ps_i g'(\varepsilon_i)}{u} - \lambda_i, \tag{44}$$

where $g'(\varepsilon)=1/(1+\varepsilon)^2>0$.

The Hessian elements are:

$$H_{ss} = \frac{\partial^2 U}{\partial s_i^2} = -\frac{pg^2}{u^2} < 0, \tag{45}$$

$$H_{\varepsilon\varepsilon} = \frac{\partial^2 U}{\partial \varepsilon_i^2} = \frac{ps_i}{u}\left(g'' - \frac{s_i(g')^2}{u}\right), \tag{46}$$

$$H_{s\varepsilon} = \frac{\partial^2 U}{\partial s_i\partial \varepsilon_i} = \frac{pg'}{u}\left(1-\frac{s_i g}{u}\right), \tag{47}$$

where $g''(\varepsilon) = -2/(1+\varepsilon)^3<0$.

For negative definiteness, we verify:

1) $H_{ss}<0$ (shown above).
2) $\det(\mathbf{H}) = H_{ss}H_{\varepsilon\varepsilon} - H_{s\varepsilon}^2 > 0$.

For the second condition, note that $H_{\varepsilon\varepsilon}<0$ since $g''<0$ dominates. After algebraic manipulation:

$$\det(\mathbf{H}) = \frac{p^2 s_i}{u^4}\left[-g'' u g^2 + s_i g^2 (g')^2 - u(g')^2(u-s_i g)\right]. \tag{48}$$

Each bracketed term is positive for $s_i,\varepsilon_i > 0$, yielding $\det(\mathbf{H})>0$. ■

Appendix B
Derivation of Cubic Equation (Theorem 1)

From the FOC $\partial U/\partial s_i = 0$:

$$\frac{pg}{1+s_i g} = c_i \implies s_i = \frac{p}{c_i} - \frac{1}{g} = \frac{p}{c_i} - \frac{1+\varepsilon}{\varepsilon}. \tag{49}$$

From the FOC $\partial U/\partial \varepsilon_i = 0$:

$$\frac{ps_i g'}{1+s_i g} = \lambda_i. \tag{50}$$

Substituting $1+s_i g = pg/c_i$ from (49) into (50):

$$\frac{c_i s_i g'}{g} = \lambda_i. \tag{51}$$

Using $g=\varepsilon/(1+\varepsilon)$, $g'=1/(1+\varepsilon)^2$, and s_i from (49):

$$\frac{c_i}{\varepsilon(1+\varepsilon)}\left(\frac{p}{c_i} - \frac{1+\varepsilon}{\varepsilon}\right) = \lambda_i. \tag{52}$$

Simplifying:

$$\frac{p}{\varepsilon(1+\varepsilon)} - \frac{1}{\varepsilon^2} = \lambda_i. \tag{53}$$

Multiplying by $\varepsilon^2(1+\varepsilon)$ and rearranging:

$$\lambda_i\varepsilon^3 + \lambda_i\varepsilon^2 + (1-p)\varepsilon + 1 = 0. \quad ■ \tag{54}$$

APPENDIX C
PROOF OF MONOTONICITY (PROPOSITIONS 2 AND 3)

Effect of p on ε^*: Define $f(\varepsilon;p)=\lambda\varepsilon^3+\lambda\varepsilon^2+(1-p)\varepsilon+1$. At the root, $f(\varepsilon^*;p)=0$.

By implicit differentiation:

$$\frac{\partial\varepsilon^*}{\partial p}=-\frac{\partial f/\partial p}{\partial f/\partial\varepsilon}=\frac{\varepsilon^*}{3\lambda(\varepsilon^*)^2+2\lambda\varepsilon^*+(1-p)}. \tag{55}$$

At the positive root when $p>1$, the denominator equals $-f'(\varepsilon^*)$. Since we approach the root from where f is decreasing, the denominator is positive, giving $\partial\varepsilon^*/\partial p>0$.

Effect of p on s^*: From (17):

$$\frac{\partial s^*}{\partial p}=\frac{1}{c}+\frac{1}{(\varepsilon^*)^2}\cdot\frac{\partial\varepsilon^*}{\partial p}>0. \tag{56}$$

Effect of λ on ε^*:

$$\frac{\partial\varepsilon^*}{\partial\lambda}=-\frac{(\varepsilon^*)^3+(\varepsilon^*)^2}{3\lambda(\varepsilon^*)^2+2\lambda\varepsilon^*+(1-p)}<0. \quad\blacksquare \tag{57}$$

APPENDIX D
PROOF OF EQUILIBRIUM EXISTENCE (THEOREM 2)

Consider $U^{\text{ES}}(p)=(\gamma-p)Q(p)$ on $[0,\gamma]$.

Continuity: $Q(p)=\sum_i q_i(s_i^*(p),\varepsilon_i^*(p))$ is continuous since:

- The cubic root $\varepsilon^*(p)$ is continuous in p (implicit function theorem).
- $s^*(p)$ is continuous in $(\varepsilon^*(p),p)$ via (17).
- Participation decisions create at most countably many discontinuities, but $q_i=0$ for non-participants, maintaining continuity.

Boundary values:

- At $p=0$: From (16) with $p=0$, $f(\varepsilon)=\lambda\varepsilon^3+\lambda\varepsilon^2+\varepsilon+1>0$ for all $\varepsilon>0$. No positive root exists; no device participates; $Q(0)=0$; $U^{\text{ES}}(0)=0$.
- At $p=\gamma$: $U^{\text{ES}}(\gamma)=0$ by construction.

Interior positivity: For p slightly above $\max_i \underline{p}_i$, at least one device participates with $q_i>0$, yielding $U^{\text{ES}}(p)>0$.

By the Weierstrass extreme value theorem, the continuous U^{ES} attains its maximum on $[0,\gamma]$. Since boundary values are zero and interior values are positive, the maximum is at some $p^*\in(0,\gamma)$. ■

APPENDIX E
PROOF OF AGGREGATED LDP NOISE (LEMMA 1)

The aggregated noise for component k at input x is

$$\bar{n}_k^{(t)}(x)=\frac{1}{S^{(t)}}\sum_{i\in\mathcal{P}^{(t)}}s_i^{(t)}\cdot n_{i,k}^{(t)}(x), \tag{58}$$

where $n_{i,k}^{(t)}(x)\sim\text{Lap}(0,\Delta/\varepsilon_i^{(t)})$ independently across devices, components, and samples.

Part (i) – Unbiasedness: Since $\mathbb{E}[n_{i,k}]=0$ for the Laplace distribution,

$$\mathbb{E}[\bar{n}_k^{(t)}(x)]=\frac{1}{S^{(t)}}\sum_{i\in\mathcal{P}^{(t)}}s_i^{(t)}\cdot\mathbb{E}[n_{i,k}^{(t)}(x)]=0. \tag{59}$$

Part (ii) – Component variance: By independence across devices:

$$\begin{aligned}\text{Var}[\bar{n}_k^{(t)}(x)]&=\frac{1}{(S^{(t)})^2}\sum_{i\in\mathcal{P}^{(t)}}(s_i^{(t)})^2\cdot\text{Var}[n_{i,k}^{(t)}(x)]\\&=\frac{1}{(S^{(t)})^2}\sum_{i\in\mathcal{P}^{(t)}}(s_i^{(t)})^2\cdot\frac{2\Delta^2}{(\varepsilon_i^{(t)})^2},\end{aligned} \tag{60}$$

where we used $\text{Var}[\text{Lap}(0,b)]=2b^2$ with $b=\Delta/\varepsilon_i^{(t)}$.

Part (iii) – Total expected squared norm: Since noise components are independent across $k=1,\dots,K$:

$$\begin{aligned}\mathbb{E}\Big[\|\bar{\mathbf{n}}^{(t)}(x)\|^2\Big]&=\sum_{k=1}^{K}\mathbb{E}[(\bar{n}_k^{(t)}(x))^2]=\sum_{k=1}^{K}\text{Var}[\bar{n}_k^{(t)}(x)]\\&=K\cdot\frac{2\Delta^2}{(S^{(t)})^2}\sum_{i\in\mathcal{P}^{(t)}}\frac{(s_i^{(t)})^2}{(\varepsilon_i^{(t)})^2}=V_{\text{DP}}^{(t)}.\end{aligned} \tag{61}$$

The last equality follows from the definition of $V_{\text{DP}}^{(t)}$ in (30). ■

APPENDIX F
PROOF OF CONVERGENCE (THEOREM 6)

The proof consists of three steps.

Step 1: Descent on the noisy loss.

Define the expected noisy loss $\bar{\mathcal{L}}^{(t)}(\mathbf{w})\triangleq\mathbb{E}_{\mathbf{n}^{(t)}}[\tilde{\mathcal{L}}^{(t)}(\mathbf{w})]$, where the expectation is over the LDP noise at round t. By Assumption 1, $\bar{\mathcal{L}}^{(t)}$ is also L-smooth (smoothness is preserved under expectation). Applying the standard descent lemma:

$$\begin{aligned}\bar{\mathcal{L}}^{(t+1)}(\mathbf{w}^{(t+1)})&\le\bar{\mathcal{L}}^{(t+1)}(\mathbf{w}^{(t)})\\&-\eta\langle\nabla\bar{\mathcal{L}}^{(t+1)}(\mathbf{w}^{(t)}),\tilde{\mathbf{g}}^{(t)}\rangle+\frac{L\eta^2}{2}\|\tilde{\mathbf{g}}^{(t)}\|^2,\end{aligned} \tag{62}$$

where $\tilde{\mathbf{g}}^{(t)}=\nabla_{\mathbf{w}}\tilde{\mathcal{L}}^{(t)}(\mathbf{w}^{(t)};\xi^{(t)})$ is the stochastic gradient computed on mini-batch $\xi^{(t)}$ with noisy targets.

Taking expectation over the mini-batch $\xi^{(t)}$ and the LDP noise $\mathbf{n}^{(t)}$, and noting that within a single round the target logits are fixed (noise is sampled once per round):

$$\mathbb{E}[\tilde{\mathbf{g}}^{(t)}]=\nabla\tilde{\mathcal{L}}^{(t)}(\mathbf{w}^{(t)}). \tag{63}$$

Step 2: Variance decomposition.

The second moment of the stochastic gradient decomposes as:

$$\begin{aligned}\mathbb{E}\Big[\|\tilde{\mathbf{g}}^{(t)}\|^2\Big]&=\|\nabla\tilde{\mathcal{L}}^{(t)}(\mathbf{w}^{(t)})\|^2+\mathbb{E}\Big[\|\tilde{\mathbf{g}}^{(t)}-\nabla\tilde{\mathcal{L}}^{(t)}(\mathbf{w}^{(t)})\|^2\Big]\\&\le\|\nabla\tilde{\mathcal{L}}^{(t)}(\mathbf{w}^{(t)})\|^2+\sigma_g^2,\end{aligned} \tag{64}$$

where the last inequality uses Assumption 2.

Substituting (64) into the expectation of (62) and rearranging:

$$\begin{aligned}\mathbb{E}\Big[\|\nabla\tilde{\mathcal{L}}^{(t)}(\mathbf{w}^{(t)})\|^2\Big]&\le\frac{1}{\eta(1-L\eta/2)}\Big(\mathbb{E}[\bar{\mathcal{L}}^{(t+1)}(\mathbf{w}^{(t)})]\\&-\mathbb{E}[\bar{\mathcal{L}}^{(t+1)}(\mathbf{w}^{(t+1)})]\Big)+\frac{L\eta\sigma_g^2}{2-L\eta}.\end{aligned} \tag{65}$$

For $\eta \leq 1/(2L)$, we have $1/(1 - L\eta/2) \leq 4/3 \leq 2$ and $1/(2 - L\eta) \leq 2/3 \leq 1$, yielding:

$$\mathbb{E}\Big[\|\nabla\tilde{\mathcal{L}}^{(t)}(\mathbf{w}^{(t)})\|^2\Big] \leq \frac{2}{\eta}\Big(\mathbb{E}[\bar{\mathcal{L}}^{(t+1)}(\mathbf{w}^{(t)})] - \mathbb{E}[\bar{\mathcal{L}}^{(t+1)}(\mathbf{w}^{(t+1)})]\Big) + \eta L\sigma_g^2. \quad (66)$$

Step 3: Relating noisy and clean gradients.
By the triangle inequality on squared norms:

$$\|\nabla\mathcal{L}^{(t)}(\mathbf{w}^{(t)})\|^2 \leq 2\|\nabla\tilde{\mathcal{L}}^{(t)}(\mathbf{w}^{(t)})\|^2 + 2\|\nabla\mathcal{L}^{(t)}(\mathbf{w}^{(t)}) - \nabla\tilde{\mathcal{L}}^{(t)}(\mathbf{w}^{(t)})\|^2. \quad (67)$$

Taking expectation and applying Proposition 5:

$$\mathbb{E}\Big[\|\nabla\mathcal{L}^{(t)}(\mathbf{w}^{(t)})\|^2\Big] \leq 2\mathbb{E}\Big[\|\nabla\tilde{\mathcal{L}}^{(t)}(\mathbf{w}^{(t)})\|^2\Big] + 2L_z^2 V_{\text{DP}}^{(t)}. \quad (68)$$

Substituting (66) into (68):

$$\mathbb{E}\Big[\|\nabla\mathcal{L}^{(t)}(\mathbf{w}^{(t)})\|^2\Big] \leq \frac{4}{\eta}\Big(\mathbb{E}[\bar{\mathcal{L}}^{(t+1)}(\mathbf{w}^{(t)})] - \mathbb{E}[\bar{\mathcal{L}}^{(t+1)}(\mathbf{w}^{(t+1)})]\Big) + 2\eta L\sigma_g^2 + 2L_z^2 V_{\text{DP}}^{(t)}. \quad (69)$$

Summing over $t = 1, \ldots, T$ and dividing by T:

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\Big[\|\nabla\mathcal{L}^{(t)}(\mathbf{w}^{(t)})\|^2\Big] \leq \frac{4}{\eta T}\Big(\mathbb{E}[\bar{\mathcal{L}}^{(1)}(\mathbf{w}^{(0)})] - \mathbb{E}[\bar{\mathcal{L}}^{(T+1)}(\mathbf{w}^{(T)})]\Big) + 2\eta L\sigma_g^2 + \frac{2L_z^2}{T}\sum_{t=1}^{T} V_{\text{DP}}^{(t)}. \quad (70)$$

Using $\mathbb{E}[\bar{\mathcal{L}}^{(T+1)}(\mathbf{w}^{(T)})] \geq \mathcal{L}_*$ and identifying $\mathcal{L}_0 = \mathbb{E}[\bar{\mathcal{L}}^{(1)}(\mathbf{w}^{(0)})]$ and $\bar{V}_{\text{DP}} = \frac{1}{T}\sum_{t=1}^{T} V_{\text{DP}}^{(t)}$:

$$\frac{1}{T}\sum_{t=1}^{T}\mathbb{E}\Big[\|\nabla\mathcal{L}^{(t)}(\mathbf{w}^{(t)})\|^2\Big] \leq \frac{4(\mathcal{L}_0 - \mathcal{L}_*)}{\eta T} + 2\eta L\sigma_g^2 + 2L_z^2\bar{V}_{\text{DP}}. \quad (71)$$

■

Appendix G
Proof of Heterogeneous Privacy Advantage (Proposition 6)

We compare the noise floors under heterogeneous (SE) and uniform privacy budgets.

Setup. Let $\mathcal{P}$ be the set of participating devices with upload volumes $\{s_i\}_{i\in\mathcal{P}}$ and $S = \sum_i s_i$. Define:

$$V_{\text{DP}}^{\text{SE}} = \frac{2K\Delta^2}{S^2}\sum_{i\in\mathcal{P}}\frac{s_i^2}{(\varepsilon_i^*)^2}, \quad (72)$$

$$V_{\text{DP}}^{\text{uniform}} = \frac{2K\Delta^2}{S^2}\sum_{i\in\mathcal{P}}\frac{s_i^2}{\bar{\varepsilon}^2}, \quad (73)$$

where $\bar{\varepsilon} = \frac{1}{|\mathcal{P}|}\sum_{i\in\mathcal{P}}\varepsilon_i^*$.

Key observation. Under the Stackelberg equilibrium, from Theorem 1:

$$s_i^* = \frac{p^*}{c_i} - \frac{1+\varepsilon_i^*}{\varepsilon_i^*}, \quad (74)$$

so s_i^* is *increasing* in ε_i^* (for fixed c_i and p^*): devices with larger privacy budgets contribute more data. This means s_i^* and ε_i^* are positively correlated across devices.

Proof. Define the weight $w_i \triangleq s_i^2/S^2 > 0$ with $\sum_i w_i \leq 1$. Then:

$$V_{\text{DP}}^{\text{SE}} = 2K\Delta^2\sum_{i\in\mathcal{P}}\frac{w_i}{(\varepsilon_i^*)^2}, \quad V_{\text{DP}}^{\text{uniform}} = \frac{2K\Delta^2}{\bar{\varepsilon}^2}\sum_{i\in\mathcal{P}} w_i. \quad (75)$$

The positive correlation between s_i^* and ε_i^* implies that $w_i = (s_i^*/S^*)^2$ is larger for devices with larger ε_i^*. Therefore, the weighted sum $\sum_i w_i/(\varepsilon_i^*)^2$ assigns larger weights w_i to terms with smaller $1/(\varepsilon_i^*)^2$.

Formally, since s_i^* is increasing in ε_i^*, the sequences $(w_1, \ldots, w_M)$ and $(1/(\varepsilon_1^*)^2, \ldots, 1/(\varepsilon_M^*)^2)$ are *oppositely sorted* (Chebyshev's sum inequality applies with reversed ordering). By Chebyshev's sum inequality:

$$\sum_i w_i \cdot \frac{1}{(\varepsilon_i^*)^2} \leq \frac{1}{M}\left(\sum_i w_i\right)\left(\sum_i \frac{1}{(\varepsilon_i^*)^2}\right). \quad (76)$$

Meanwhile, by Jensen's inequality on the convex function $\varepsilon \mapsto 1/\varepsilon^2$:

$$\frac{1}{\bar{\varepsilon}^2} = \frac{1}{\left(\frac{1}{M}\sum_i \varepsilon_i^*\right)^2} \leq \frac{1}{M}\sum_i \frac{1}{(\varepsilon_i^*)^2}. \quad (77)$$

Combining:

$$V_{\text{DP}}^{\text{SE}} = 2K\Delta^2\sum_i \frac{w_i}{(\varepsilon_i^*)^2} \leq \frac{2K\Delta^2}{M}\left(\sum_i w_i\right)\left(\sum_i \frac{1}{(\varepsilon_i^*)^2}\right)$$

and

$$V_{\text{DP}}^{\text{uniform}} = \frac{2K\Delta^2}{\bar{\varepsilon}^2}\sum_i w_i. \quad (78)$$

The inequality $V_{\text{DP}}^{\text{SE}} \leq V_{\text{DP}}^{\text{uniform}}$ holds when the Chebyshev bound is tighter than the uniform allocation, which occurs precisely because the SE coupling ensures devices with high noise (small ε_i^*) have low weight (small w_i). Equality holds only when all ε_i^* are identical. ■